

If G is a connected, undirected graph, then the largest eigenvalue of W_G , ω_1 , has multiplicity 1, equals 1, and has eigenvector $\mathbf{d}$. We may convert this eigenvector into a probability distribution $\boldsymbol{\pi}$ by setting

$$\boldsymbol{\pi} = \frac{\mathbf{d}}{\sum_a \mathbf{d}(a)}.$$

If $\omega_n \neq -1$, then the distribution of every random walk eventually converges to $\boldsymbol{\pi}$. The rate of this convergence is governed by how close $\max(\omega_2, -\omega_n)$ is to ω_1 . For example, let $\mathbf{p}_t$ denote the distribution after t steps of a random walk that starts at vertex a . Then for every vertex b ,

$$|\mathbf{p}_t(b) - \boldsymbol{\pi}(b)| \leq \sqrt{\frac{\mathbf{d}(b)}{\mathbf{d}(a)}} (1 - \max(\omega_2, -\omega_n))^t.$$